Rebecca Ann Jackson

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EDUCATION

Villanova University – Villanova, PA

- M.S. in Computer Science | GPA: 4.00 — Aug 2026 – May 2027
- B.S. in Computer Science & B.S. in Mathematics | GPA: 3.99 — Aug 2023 – May 2026
- **Honors:** Phi Beta Kappa (2025), Upsilon Pi Epsilon (2026), Pi Mu Epsilon (2026)

WORK EXPERIENCE

Graduate Assistant | Villanova University — Aug 2026 – Present

- Teaching assistant for Statistics in Computing and Agentic AI: taught R programming class sessions and graded data analysis and agentic LLM application projects (MCP, LangChain) for ~50 students
- Lead ongoing research on Carbon Calculus, extending its emissions and lifecycle cost models (see Projects)

Business Insights & Analytics Intern | Wawa Corporate Headquarters, Media, PA — May – Aug 2026

- Built LightGBM models forecasting weekly store-level sales 6 periods out for budgeting, reducing WMAPE from 20% to 5% through feature engineering and model tuning
- Queried and analyzed 50M+ rows of transaction/customer data with SQL in Snowflake to support Financial Planning & Analysis
- Developed Tableau and Excel dashboards tracking hourly sales for C-suite, used by leadership to monitor daypart performance

Undergraduate Research Assistant | Villanova University — Jan 2024 – Aug 2026

- Contributed to 4 projects with Python data analysis, API integration, and web development; co-authored 2 peer-reviewed publications:
  - *#Haul: Analyzing YouTube Discourse on Over-Consumption & Sustainability* (NordiCHI 2024): pulled 1,000+ videos and engagement metrics via the YouTube Data API in Python, cleaned the data to 708 records, and ran a mixed-methods analysis
  - *Non-Recursive Solution of the Tower of Hanoi* (Mathematics Magazine, 2025): co-derived a closed-form equation that gives the puzzle state at any move without recursion, and built an interactive JavaScript visualization of the solution

Undergraduate Teaching Assistant | Villanova University — Jan 2025 – May 2026

- Teaching Assistant for Competitive Programming and Data Structures & Algorithms: gave detailed assignment feedback to 30+ students, held weekly office hours, and ran one-on-one mock coding interviews

PROJECTS

Carbon Calculus – JavaScript, React, Recharts, Supabase, Vercel, Vite — Jun 2025 – Present

- Built and deployed an interactive tool on Vercel that models CO_2 reductions and lifecycle costs, computing NPV across a case study → scenario → project data hierarchy stored in Supabase
- Designed dynamic visualizations (cumulative and stacked comparisons) and validated JSON import/export so users can compare scenarios
- First undergraduate project funded by Villanova's Strategic Initiative for Climate, Justice, and Sustainability

American Sign Language Classifier – Python, PyTorch, Scikit-Learn — Aug – Dec 2025

- Trained a CNN to classify 36 ASL letters and digits from 2,515 images, reaching 95% accuracy on a held-out test set of 378 images after tuning and comparing 5 architectures
- Analyzed per-class precision and recall to trace errors to visually similar signs (M vs. N, W vs. 6); published a reproducible end-to-end pipeline on GitHub

YouTube Video Perception Timeline – React, TypeScript, YouTube Data API — Oct 2026 – Present

- Building an app with a 2-person team that tracks how a video's audience reception changes over time by combining YouTube engagement metrics with comment sentiment analysis
- Designing a pipeline that pulls comments and metrics through the YouTube Data API, scores sentiment with a pre-trained model, and visualizes trends on an interactive timeline

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, JavaScript, TypeScript, HTML/CSS, Java, C/C++, LaTeX
- **Analytics & BI:** Tableau, Snowflake, Excel
- **Data & ML:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, LightGBM, Jupyter, PyTorch
- **Methods:** Data Cleaning, Data Visualization, Statistical Modeling, Time-Series Forecasting, Machine Learning, Sentiment Analysis

LEADERSHIP AND ACTIVITIES

President, Villanova Programming Team — Sep 2024 – Present

- Lead weekly competitive programming practice sessions for 130+ members in Python, C++, and Java
- Led team to top ICPC regional finishes: 1st place at competition site in Division 2 (2024) and 2nd place in Division 1 (2025)

President, Girls Who Code Villanova — Apr 2025 – Present

- Lead a community of 50+ women through educational, social, and technical meetings and events